MIL–F–11245A (MU)
15 OCTOBER 1963
SUPERSEDING
MIL–F–11245 (Ord)
29 JUNE 1951

MILITARY SPECIFICATION

FUZE, MINE, AT, M603
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of fuze designated as Fuze, Mine, AT, M603.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ–A–561 —Aluminum Alloy 1100 (2S); Plate and Sheet

MILITARY

MIL–D–2498 —Detonators, Loading, Assembling and Packing

MIL–A–2550 —Ammunition and Special Weapons, General Specification for

MIL–I–45208 —Inspection Requirements, General Specification for

MIL–I–45607 —Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes

MIL–STD–109 —Quality Assurance Terms and Definitions

MIL–STD–1235 —Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

73–9–55 —Fuze, Mine, AT, M603, Assembly

PUBLICATIONS

ORDNANCE CORPS

IEL—7300427 —Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of

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invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71–90 —Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71–90 (latest revisions)".)

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Fuze assemblies. The fuze assemblies shall comply with all requirements specified on Drawing (dwg.) 73–9–55 and with all requirements specified in applicable specifications.

3.3 Detonator crimp. The crimp of the detonator to the body shall withstand a 30 pound load without visible distortion or failure when tested as specified in 4.4.1.

3.4 Waterproofness and functioning. The fuzes shall show no evidence of leakage and shall detonate high order (see 6.4) when a load of 140 to 240 pounds is applied, after temperature conditioning and submergence in water as specified in 4.4.2. The fuzes shall not function when a load of 135 pounds is applied.

3.5 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

(b) Number of units of product inspected.

(c) Results obtained for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of analyses on all material purchased by the contractor

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when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection.

4.2.1 *Submission.* Prior to initiation of normal production the contractor shall submit an initial production sample consisting of 50 consecutively produced assemblies which have been produced by the production process which the contractor will use in fulfilling the contract, to a Government-approved facility designated by the contracting officer. All parts and materials shall be obtained from the same source of supply and shall be the same as will be used in regular production. The initial production sample shall be accompanied by certificates of analysis for all materials. A sample shall also be submitted whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.2 *Inspections to be performed.* Each assembly and component shall be inspected by the Government for all the inspections specified in 4.3.2 and 4.3.3 of this specification and for all additional requirements of the applicable drawings.

4.2.3 *Rejection.* If any assembly fails to comply with the applicable requirements, the initial production sample shall be rejected. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit new or supplemental quantities until such time as an acceptable sample is submitted, or until otherwise directed by the contracting officer.

4.3 Inspection provisions.

4.3.1 *Lot formation.* A lot shall consist of fuze assemblies produced by one manufacture in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot number. Each lot shall contain:

(a) Parts and subassemblies from one interfix lot number from one manufacturer.

(b) Detonators from one interfix lot number from one manufacturer.

4.3.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect

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and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Assembly (see dwg. 73-9-55).

Categories	Defects	Method of inspection	Code No. (see 6.2)
Critical:			
1.	Safety fork missing or loose	Visual-Manual	01001
Major:	AQL 0.65 percent		
101.	Distance from detonator well shoulder to firing pin tip (before assembling detonator)	Gage	01002
102.	Total length	Gage	01003
103.	Detonator above flush	Gage	01004
104.	Detonator inverted	Visual	01005
105.	Detonator crimp fractured or not for 360 degrees	Visual	01006
106.	Lining compound missing or inadequate	Visual	01007
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	01008

4.3.2.2 Carton, sealed (see section 5).

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Evidence of loose contents	Manual	02001
102.	Carton damaged so that contents are exposed or liable to become exposed	Visual	02002
103.	Closing for shipment improper	Visual	02003
Minor:	AQL 0.40 percent		
201.	Marking misleading or improper	Visual	02004

4.3.3 *Testing.*

4.3.3.1 Detonator crimp (see 3.3) major defect. A sample of 30 assemblies shall be selected for this test. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4. If two or more samples do not comply with 3.3 the lot shall be rejected. If one sample does not comply with 3.3 the lot shall be rejected unless the supplier requests, and the lot passes, a retest. A sample of 60 additional assemblies shall be selected for the retest and failure of one assembly during the retesting shall cause the lot to be rejected.

01010

4.3.3.2 Waterproofness and Functioning (see 3.4) Critical defect if functioning load is too low; major defect for evidence of leakage or if functioning load is too high. A sample of 30 fuzes shall be selected from each lot for this test. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4. If one or more samples do not comply with 3.4 the lot shall be rejected.

03001

4.3.3.3 Check test for deterioration of detonators (see Specification MIL-D-2493) Major defect. If the total time between original acceptance of any detonator lot and assembly of that lot into the mine fuze exceeds two years or if the detonators have been subjected to adverse conditions, however brief, at any time since previous tests, the detonator lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-D-2493 immediately before the lot is assembled into the fuzes. This test shall be performed by the contractor on detonators selected by the Government inspector at the facility assembling the detonators into the fuzes. The test shall be conducted at Government expense, without cost to the contractor who loaded the detonator, or to the contractor assembling the detonators into the fuzes, and shall not constitute a basis of rejection against either

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contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc. permitted while the lot was under jurisdiction of either contractor. (see 6.5)

4.3.4 *Inspection equipment.* Index of Inspection Equipment Number IEL 7800427 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.3.4.3 *Government use of contractor's inspection equipment.* The contractor shall as required make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Operation of the equipment and verification of its accuracy shall be supplied by the contractor, as required by the Government in performance of its examination or test.

4.4 **Test methods and procedures.**

4.4.1 *Detonator crimp.* The cover assembly and the firing pin assembly shall be removed from the fuze assembly. The lower portion of the body shall be securely supported with the detonator well fully exposed. A load of 30 pounds shall be applied to the upper end of the detonator, at such a rate that the load fixture moves at a rate of ½ to 1 inch per minute. Observation shall be made for no movement of the lower end of the detonator and no deformation or fracturing of the crimp to determine compliance with 3.3. (Note: This test should be conducted behind a suitable barricade.)

4.4.2 *Waterproofness and functioning.* The fuze assemblies selected shall be subjected to 4 conditioning cycles, each consisting of storage for at least 4 hours at minus 65 plus or minus 2 degrees Fahrenheit followed by storage for at least 16 hours at 160 plus or minus 2 degrees Fahrenheit and 95 percent relative humidity. The fuzes shall then be submerged in 6 to 8 inches of water at 40 plus or minus 2 degrees Fahrenheit and fuzes and water shall be allowed to return to room temperature (70 plus or minus 10 degrees Fahrenheit). Total submersion time shall be at least 24 hours. Within one-half hour after their removal from the water 10 of the fuzes shall be dismantled and examined visually for evidence of leakage and moisture to determine compliance with 3.4. The remaining 20 fuzes shall be tested for functioning as follows. The lower portion of the body shall be supported securely and the safety fork removed. A load shall be applied to the pressure plate at such a rate that the load fixture moves at a rate of ½ to 1 inch per minute. Observation shall be made for no functioning at a load of 135 pounds and for high order detonation at a load of 140 to 240 pounds to determine compliance with 3.4.

5. PREPARATION FOR DELIVERY

5.1 **Preservation and packaging.**

5.1.1 *Level C.* Packaging for Fuze, Mine, AT, M603 shall be in accordance with ICC Regulation 49 CFR 71-78 in such a manner that no damage will occur during shipment.

5.2 **Packing.**

5.2.1 *Level C.* Packing for Fuze, Mine, AT, M603 shall be in accordance with ICC Regulation 49 CFR 71-78.

5.3 **Data cards.** Data card information shall be as specified in Specification MIL-A-2550.

5.4 **Marking.** Marking of the cartons shall

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be in accordance with ICC Regulations 49 CFR 71–78 and Standard MIL–STD–129.

5.5 **Shipping.** When fuze assemblies from more than one lot are shipped as a carload, each lot must be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 **Ordering data.** Procurement documents shall specify the following:

(a) Title, number and date of this specification.

6.2 **Inspection code numbers.** The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 **Inspection equipment.** The contractor will design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF —Contractor's design responsibility on Ordnance format; ORDM–608–12.
CDCF —Contractor's design responsibility on contractor format.
OD —Ordnance design.
ODMU —Ordnance design, mandatory for use.
CE —Commercial equipment.
SCD —Specification Control Drawing.

6.3.3 *Ordnance designs.* Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawings.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation

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by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial equipment but a list of such equipment must be approved by the Government. Lists will be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608–12, and, in addition, contain the detail and information specified above.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 The criterion of high order detonation shall be penetration of a disc securely clamped to the fuze over the exposed end of the fuze detonator. The disc shall be 1.218 minus 0.010 inches in diameter and 0.1250 plus or minus 0.0045 inches thick and shall be made of aluminum in accordance with Specification QQ–A–561.

6.5 **Check test.** The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of detonators.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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